

I. IDENTIFICATION DATA

Thesis title:	Transfer learning for cloud semantic segmentation on satellite imagery
Author's name:	Bc. Matěj Klimeš
Type of thesis :	master
Faculty/Institute:	Faculty of Civil Engineering (FCE)
Department:	Department of Geomatics
Thesis reviewer:	Prof. Arnon Karnieli, Ph.D.
Reviewer's department:	Click here to enter text.

II. EVALUATION OF INDIVIDUAL CRITERIA

Assignment	challenging
<i>How demanding was the assigned project?</i>	
The project presented a high level of technical and operational demand. It required ingesting, preprocessing, and calibrating two distinct satellite imagery datasets: VENμS and Sentinel-2. Following data preparation, the deep learning model was trained for cloud detection and benchmarked against reference data. This evaluation process involved extensive tuning and the implementation of various data normalization techniques. Consequently, the comprehensive nature of the data integration and modeling pipeline justifies classifying this project as a challenge.	

Fulfilment of assignment	fulfilled
<i>How well does the thesis fulfil the assigned task? Have the primary goals been achieved? Which assigned tasks have been incompletely covered, and which parts of the thesis are overextended? Justify your answer.</i>	
The thesis fully answers the research questions, and the primary goals identified have been achieved.	

Methodology	outstanding
<i>Comment on the correctness of the approach and/or the solution methods.</i>	
The chosen methodology is robust and well-supported by established academic literature. The approach is efficient for demonstrating cross-sensor transfer learning within the context of cloud detection. While more recent or architecturally complex deep learning models are present in the current literature, the U-Net architecture remains highly appropriate. It serves as a reliable benchmark and demonstrates significant potential.	

Technical level	A - excellent.
<i>Is the thesis technically sound? How well did the student employ expertise in the field of his/her field of study? Does the student explain clearly what he/she has done?</i>	
The technical execution of the thesis is of a high standard. The manuscript provides a comprehensive discussion of the theoretical and scientific background, drawing on contemporary methodologies in transfer learning. The document demonstrates structural clarity, thus bridging the complexities of both the remote sensing and machine learning components of the research.	

Formal and language level, scope of thesis	A - excellent.
<i>Are formalisms and notations used properly? Is the thesis organized in a logical way? Is the thesis sufficiently extensive? Is the thesis well-presented? Is the language clear and understandable? Is the English satisfactory?</i>	
The thesis satisfies all formal, structural, and linguistic requirements. Mathematical formalisms and notations are applied correctly and consistently throughout the text. The manuscript is logically organized, maintaining a coherent progression from the theoretical framework and data preprocessing stages to the methodological implementation and the final discussion.	

Selection of sources, citation correctness

A - excellent.

Does the thesis make adequate reference to earlier work on the topic? Was the selection of sources adequate? Is the student's original work clearly distinguished from earlier work in the field? Do the bibliographic citations meet the standards?

The thesis makes comprehensive and adequate reference to earlier work, demonstrating a thorough understanding of the research landscape.

Additional commentary and evaluation (optional)

Comment on the overall quality of the thesis, its novelty and its impact on the field, its strengths and weaknesses, the utility of the solution that is presented, the theoretical/formal level, the student's skillfulness, etc.

I thoroughly enjoyed reading this thesis. It is clear, highly engaging, and tackles a very practical challenge in remote sensing. Successfully implementing cross-sensor transfer learning for cloud detection using Sentinel-2 and Venus imagery offers valuable, realistic insights for operational workflows. The results are highly interesting and backed by a robust benchmarking process.

III. OVERALL EVALUATION, QUESTIONS FOR THE PRESENTATION AND DEFENSE OF THE THESIS, SUGGESTED GRADE

Summarize your opinion on the thesis and explain your final grading. Pose questions that should be answered during the presentation and defense of the student's work.

The grade that I award for the thesis is A - excellent.

Date: **Click here and enter the date.**

June 16, 2026

Signature:

